



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
21-02-2021

JEE(Main+Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A), I, LIVE-I(A) & LIVE-I]

PAPER-2

Time : 3 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

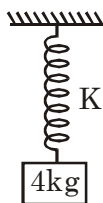
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. A sphere of mass m and radius r is projected in a gravity free space with speed v . If coefficient of viscosity of the medium in which it moves is $\frac{1}{6\pi}$, the distance travelled by the body before it stops is :
- (A) $\frac{mv}{2r}$ (B) $\frac{2mv}{r}$ (C) $\frac{mv}{r}$ (D) $\frac{mv}{4r}$
2. A body of mass 4kg is suspended from a spring of spring constant 400 N/m . Another body of mass 4 kg moving vertically upward with 2 m/s hits it & gets embedded in it. Amplitude of motion is :



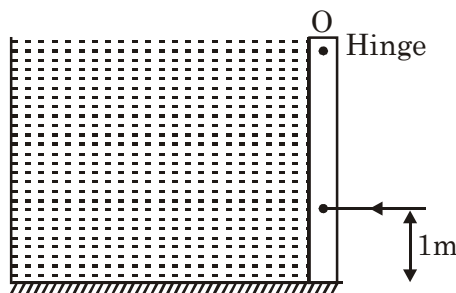
- (A) $\sqrt{\frac{3}{100}}\text{ m}$ (B) $\sqrt{\frac{3}{200}}\text{ m}$ (C) $\sqrt{\frac{6}{100}}\text{ m}$ (D) $\sqrt{\frac{9}{200}}\text{ m}$

Space for Rough Work

3. Time period of a particle executing linear S.H.M. is $T = 4\pi$ sec. There is a point P at a distance $x = 2$ cm from the mean position O. When the particle passes through the point P towards O it has speed $v = 1$ cm/sec. The minimum time after which it will again return to point P is :

(A) $\left(\frac{\pi}{4}\right)$ sec (B) 3π sec (C) $\left(\frac{3\pi}{4}\right)$ sec (D) 2π sec

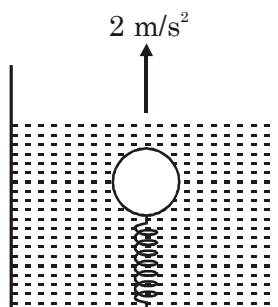
4. A square gate of size $4\text{m} \times 4\text{m}$ is hinged at topmost point. A fluid of density ρ fills the space left of it. The force acting 1m from lowest point which can hold the gate stationary is :



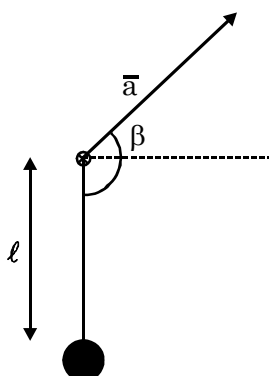
(A) $\frac{256}{3} \rho g$ (B) $\frac{256}{9} \rho g$ (C) $\frac{128}{9} \rho g$ (D) $\frac{128}{3} \rho g$

Space for Rough Work

5. A ball of mass 10kg and density 1 gm/cm^3 is attached to the base of a container having a liquid of density 1.1 gm/cm^3 , with the help of a spring as shown in the figure. The container is going up with an acceleration 2 m/s^2 . If the spring constant of the spring is 200 N/m , the elongation in the spring is :-



- (A) 2cm (B) 4 cm (C) 6 cm (D) 8 cm
6. Find the period of small oscillation of a simple pendulum of length ' ℓ ' if its point of suspension O moves relative to the earth with a constant acceleration $\vec{a}$, given $\ell = 30\text{ cm}$ and $|\vec{a}| = \frac{g}{2}$ and the angle between the vectors $\vec{g}$ and $\vec{a}$ is $\beta = 120^\circ$. The period of oscillation of the pendulum is approximately :



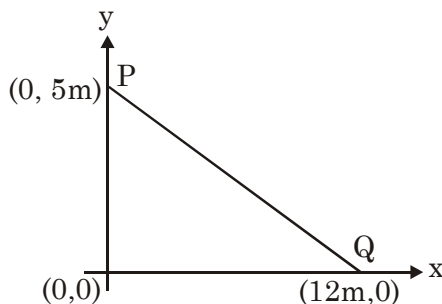
- (A) 0.5 sec (B) 4 sec (C) 0.95 sec (D) 2 sec

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

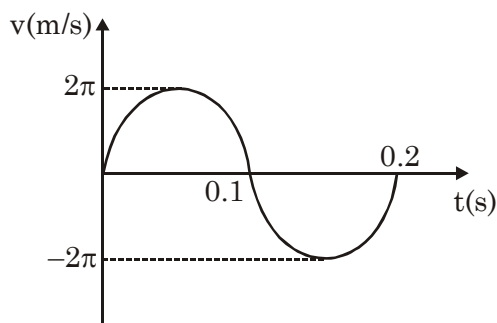
7. A particle is executing simple harmonic motion along a line $12y + 5x = 60$ only in between the extreme point 'P' and 'Q'. At $t = 0$, the particle starts its motion from the extreme point 'P'. The time period of simple harmonic motion executed by the particle is 8 sec. Then choose the **CORRECT** option(s) :



- (A) The position vector of the particle at time t is given by $\vec{r} = \left[12 \sin^2 \left(\frac{\pi t}{8} \right) \hat{i} + 5 \cos^2 \left(\frac{\pi t}{8} \right) \hat{j} \right] \text{m}.$
- (B) The position vector of the particle at time t is given by $\vec{r} = \left[12 \cos^2 \left(\frac{\pi t}{8} \right) \hat{i} + 5 \sin^2 \left(\frac{\pi t}{8} \right) \hat{j} \right] \text{m}.$
- (C) The velocity vector of the particle at time t is given by $\vec{v} = \left[\frac{3\pi}{2} \sin \left(\frac{\pi t}{4} \right) \hat{i} - \frac{5\pi}{8} \sin \left(\frac{\pi t}{4} \right) \hat{j} \right] \text{m/s}.$
- (D) The amplitude of simple harmonic motion executed by the particle is $\frac{13}{2} \text{m}.$

Space for Rough Work

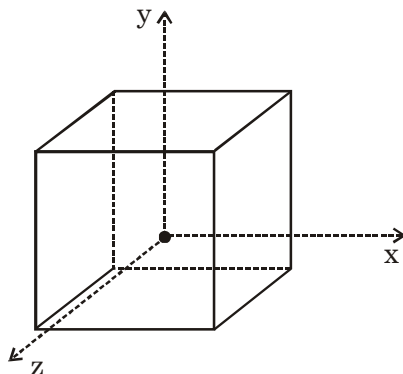
8. A spring block system performs simple harmonic motion about its equilibrium position along x-axis. The spring constant of used spring is 200 N/m and equilibrium position of block is origin. The velocity time graph of the particle is given in the figure. Choose the **CORRECT** option(s): (Take : $\pi^2 = 10$)



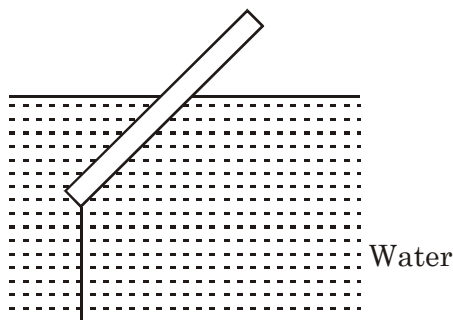
- (A) The mass of the block is 0.2 kg.
 (B) The displacement of the particle from equilibrium position at $t = 0$ is 0.1 meter.
 (C) The acceleration of block at $t = 0.1$ second is -200 m/s^2 .
 (D) The maximum kinetic energy of the block is 4.0 Joule.

Space for Rough Work

9. A rubber ball is kept at rest at the centre of a cube filled with liquid. The density of liquid is ρ and rubber ball is $\rho/3$. At $t = 0$, $\vec{a} = \left(\frac{5}{\sqrt{2}}\hat{i} - \frac{15}{2}\hat{j} + \frac{5}{2}\hat{k} \right) \text{ m/s}^2$ is the acceleration of cube and the ball is released from origin as shown in the figure. The acceleration due to gravity is along negative y-axis. At $t = 0$, choose the correct option(s) : (Take : $g = 10 \text{ m/s}^2$)



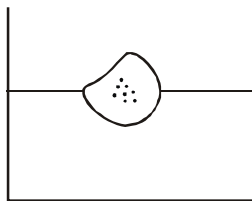
- (A) Angle made by acceleration of ball with x-axis is $\frac{\pi}{4}$ as seen from the cube.
 (B) Angle made by acceleration of ball with y-axis is $\frac{\pi}{3}$ as seen from the cube.
 (C) Angle made by acceleration of ball with z-axis is $\frac{\pi}{3}$ as seen from the cube.
 (D) If density of rubber ball is considered as $\frac{\rho}{2}$ then angle made by acceleration vector of ball with x, y and z-axis is changed as seen from the cube.
10. A stick is tied to the floor of the water tank with a string as shown in the figure. The length of stick is 2m. Area of cross-section of stick is 10^{-3} m^2 . Specific gravity of stick is 0.25. Take : $g = 10 \text{ m/s}^2$. Then :



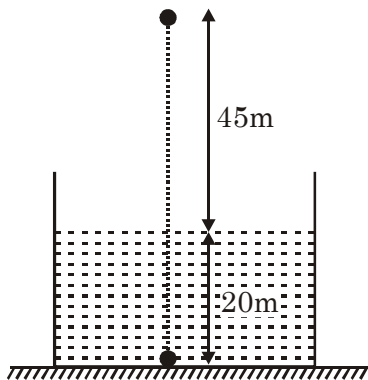
- (A) Tension in the string is 5N
 (B) Buoyancy force acting on stick is 10 N
 (C) Length of stick immersed in water is 1m
 (D) When string is cut, initial acceleration of centre of mass of stick is 10 m/s^2

Space for Rough Work

11. Some pieces of impurity (density $=\rho$) is embedded in ice. This ice is floating in water. (density $=\rho_w$). When ice melts, level of water will



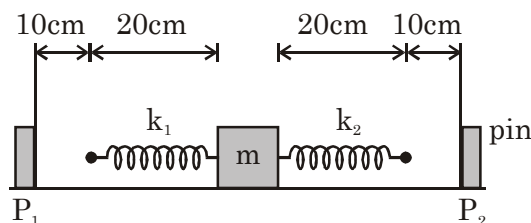
- (A) fall if $\rho > \rho_w$ (B) remain unchanged, if $\rho < \rho_w$
 (C) fall if $\rho < \rho_w$ (D) rise if $\rho > \rho_w$
12. A target object is placed at the bottom of a tank of depth 20m, and filled with a liquid of density 500 gm/cc. Another object of density 100 gm/cc is dropped onto it, from a height of 45m from the upper surface of the liquid. (take : $g = 10 \text{ m/s}^2$) (Neglect viscous force)



- (A) The second object will collide with the target.
 (B) The second object does not collide with the target.
 (C) The least separation between object and target is 11.25 m.
 (D) The retardation of the object in the liquid is 40 ms^{-2} .

Space for Rough Work

13. Two spring, each of unstretched length 20 cm but having different spring constants $k_1 = 1000 \text{ N/m}$ and $k_2 = 3000 \text{ N/m}$ are attached to two opposite faces of a block of mass $m = 100 \text{ g}$ kept on a smooth horizontal surface as shown in the figure. The outer ends of the two springs are now attached to two pins P_1 and P_2 whose locations are shown in the figure, as a result of this, the block acquires a new equilibrium position. The block has been displaced by small amount from its equilibrium position and released to perform simple harmonic motion, then:



- (A) New equilibrium position is at 35 cm from P_1 and time period of simple harmonic motion is $\frac{\pi}{100} \text{ s}$
- (B) New equilibrium position is at 20 cm from P_1 and time period of simple harmonic motion is $\frac{\pi}{100} \text{ s}$
- (C) The potential energy stored in spring of spring constant k_2 in equilibrium position is 3.75 J.
- (D) The potential energy stored in spring of spring constant k_1 in equilibrium position is 11.25 J.
14. In simple harmonic motion of amplitude A
- (A) Time taken by particle to move from mean position to $\frac{A}{2}$ is equal to time taken by particle to move from $\frac{A}{2}$ to A .
- (B) Average velocity of particle in one time period T is $\frac{4A}{T}$.
- (C) Net work done on block in one time period T is 0.
- (D) Particle has maximum kinetic energy at mean position.

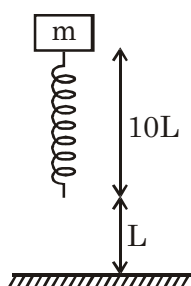
Space for Rough Work

SECTION-I(iii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

Paragraph for Questions 15 and 16

A small block of mass m is fixed at upper end of a massless vertical spring of spring constant $K = \frac{4mg}{L}$ and natural length ' $10L$ '. The lower end of spring is free and is at a height L from fixed horizontal floor as shown. The spring is initially unstretched and the spring-block system is released from rest in the shown position.



Horizontal floor

15. Till the block reaches its lowest position for the first time, the time duration for which the spring remains compressed is :

- (A) $\pi\sqrt{\frac{L}{2g}} + \sqrt{\frac{L}{4g}} \sin^{-1} \frac{3}{5}$ (B) $\frac{\pi}{4}\sqrt{\frac{L}{g}} + \sqrt{\frac{L}{4g}} \sin^{-1} \frac{1}{3}$
 (C) $\pi\sqrt{\frac{L}{2g}} + \sqrt{\frac{L}{4g}} \sin^{-1} \frac{2}{3}$ (D) $\frac{\pi}{2}\sqrt{\frac{L}{2g}} + \sqrt{\frac{L}{4g}} \sin^{-1} \frac{2}{3}$

16. When spring just touches the ground, take that instant as $t = 0$ and velocity of block at that instant as v . Then the time ' t ' at which block will have the same magnitude of velocity ' v ' is given by :

Given :

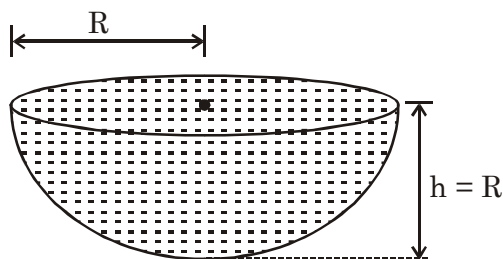
- (1) At time instant t_0 , block first time reaches its mean position.
 (2) T = time period of S.H.M. of spring block system.

- (A) $t = 3t_0$ (B) $t = \frac{T}{4}$ (C) $t = \frac{T}{2} + 2t_0$ (D) $t = \frac{T}{2} - t_0$

Space for Rough Work

Paragraph for Questions 17 and 18

A hemispherical tank of radius R is completely filled with water. Now an orifice of small area 'a' is made at the bottom of tank.



17. Time rate of volume emission from orifice when tank have $\frac{h}{2}$ height of water level is :

(A) $a\sqrt{4gh}$ (B) $a\sqrt{2gh}$ (C) $a\sqrt{gh}$ (D) $a\sqrt{\frac{gh}{2}}$

18. The time required to empty the tank is :

(A) $\frac{14\pi R^{5/2}}{15\sqrt{2} a\sqrt{g}}$ (B) $\frac{14\pi R^{5/2}}{\sqrt{2} a\sqrt{g}}$ (C) $\frac{14\pi R^{5/2}}{5\sqrt{2} a\sqrt{g}}$ (D) $\frac{4\pi R^{5/2}}{13\sqrt{2} a\sqrt{g}}$

Space for Rough Work

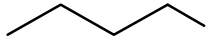
PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
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- Maximum number of atom in one plane in $\text{N}(\text{SiH}_3)_3$ is :-
 (A) 3 (B) 4 (C) 7 (D) 6
- Correct order of stability of resonating structures is :
 (i) $\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{NH}_2$ (ii) $\text{CH}_3 - \overset{\text{O}^\oplus}{\parallel} \text{C} - \text{NH}_2$ (iii) $\text{CH}_3 - \overset{\text{O}^\ominus}{\parallel} \text{C} = \overset{\oplus}{\text{N}}\text{H}_2$ (iv) $\text{CH}_3 - \overset{\text{O}^\ominus}{\parallel} \text{C}^\oplus - \text{NH}_2$
 (A) i > iii > ii > iv (B) i > ii > iii > iv (C) i > iii > iv > ii (D) i > iv > ii > iii
- Ionic compound AB has NaCl type structure. It's density is 3.6 gm/cc. When 4 gm of this compound is dissolved in 200 gm of water, one molal solution is formed. If radius of anion is $\left(\frac{302}{3}\right)$ pm and the radius of cation is x pm, then the value of $\left(\frac{x}{22}\right)$ is. ($N_A = 6 \times 10^{23}$)
 (A) 3 (B) 3.5 (C) 4 (D) 4.5

Space for Rough Work

4. Identify the process which is exothermic.
- (A) $\text{Xe} + \text{e}^- \longrightarrow$ (B) $\text{Na}^+ + \text{e}^- \longrightarrow$
 (C) $\text{S}^- + \text{e}^- \longrightarrow$ (D) $\text{Cl}^- - \text{e}^- \longrightarrow$
5. Select the **CORRECT** statement -
- (A) C_3H_7 – has four alkyl groups (structurally)
 (B) C_5H_{11} – has seven alkyl groups (structurally)
 (C) C_4H_9 – has four alkyl groups (structurally)
 (D) $\text{CH}_3 - \underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}} - \text{NH}_2$ &  NH_2 are homologues (structurally)
6. A_2B has antifluorite structure. If all ions along any one body diagonal are removed, then new formula of compound will be -
- (A) A_4B_5 (B) A_8B_5 (C) A_7B_6 (D) A_8B_4

Space for Rough Work

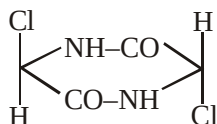
SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
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7. Which of the following species can **NOT** effectively dimerises?

- (A) BF_3 (B) AlF_3 (C) AlCl_3 (D) $\cdot\text{CF}_3$

8. Which of the following statement is/are correct about the given compound :-



- (A) It has centre of symmetry (B) It has plane of symmetry
 (C) It is optically inactive (D) It can show optical isomerism

Space for Rough Work

9. 20 mL of 34% w/v AgNO_3 aq. solution (density = 1.5 gm/mL) is mixed with 80 mL of 10% w/v CaBr_2 aq. solution. [density = 1.25 gm/mL], to form AgBr precipitate. If contraction in volume is 5% then select the **CORRECT** option(s) about resultant solution -
[Atomic mass : Ag = 108, Ca = 40, Br = 80]

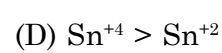
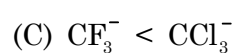
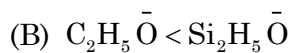
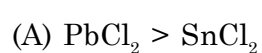
(A) % w/v of $\text{Br}^- = \frac{3.2}{95} \times 100$

(B) % w/w of $\text{Ca}^{+2} = \frac{1.6}{130} \times 100$

(C) % w/w of $\text{Br}^- = \frac{3.2}{122.48} \times 100$

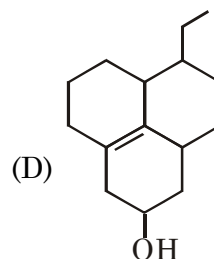
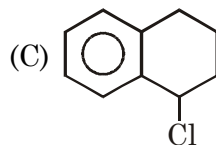
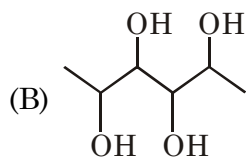
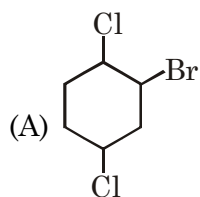
(D) % w/w of $\text{NO}_3^- = \frac{1.24}{95} \times 100$

10. Identify the correct order(s) of stability.



Space for Rough Work

11. Which of following compound has even number of chiral centre?



12. Select the correct statement(s)

- (A) Cubic crystal system has maximum symmetry while triclinic system has least symmetry.
- (B) Amorphous solids are anisotropic while crystalline solids are isotropic
- (C) Monoclinic and orthorhombic crystal systems both have end centered unit cell.
- (D) MgO and Pt both have high melting points but MgO is brittle while Pt is ductile.

13. Which of the following molecule(s) having A–A linkage (A = central atom S/Ga/N)?

- (A) $S_2O_5^{2-}$
- (B) Ga_2Cl_6
- (C) N_2O_4
- (D) S_3O_9

14. Which of the following statement is/are **CORRECT** for $BeCl_2$ (solid)?

- (A) Be is sp hybridised
- (B) Exist in polymeric form
- (C) Be having complete octet
- (D) Having (3c–4e) bond

Space for Rough Work

SECTION-I(iii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

Paragraph for Questions 15 and 16

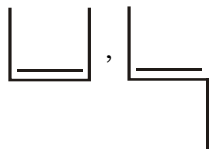
A hypothetical metal M (Atomic weight = 450) crystallizes as a cubic crystal such that metal atoms touch each other along body diagonal of cube. Edge length of cubic unit cell is 1nm. If this metal is melted then its density in molten form is 1.2 gm/cc. ($N_A = 6 \times 10^{23}$)

15. Density of solid metal in gm/cc is -
(A) 3 (B) 1.6 (C) 1.5 (D) 0.75
16. Percentage of empty space in molten state of metal will be -
(A) 20% (B) 25.6 % (C) 45.6% (D) 40.8%

Space for Rough Work

Paragraph for Questions 17 and 18

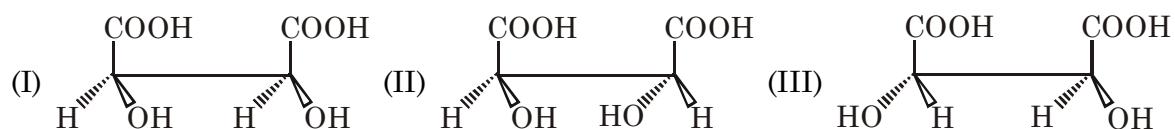
Diastereomer are a type of stereoisomers diastereomism occurs when two or more stereoisomers of a compounds have different configurations at one or more (but not all) of the relative stereocenters and are not mirror image of each other example



are diastereomers. A meso compound or meso isomer is stereoisomer with an

identical or super imposable mirror image i.e. a optically inactive member of a set of stereoisomers at least two of which one optically active this means that despite containing two or more stereogenic centre, the molecule is not chiral.

For example—Tartaric acid exist in three different optical isomers



17. Which of the following represent meso compound :
 (A) I (B) II (C) III (D) None
18. II and III are related as :
 (A) Identical compounds (B) Chain isomers
 (C) Enantiomers (D) Diastereomers

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks : 18)**

- This section contains **SIX** questions.
 - Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
 - For each question, darken the bubble corresponding to the correct option in the ORS.
 - For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases
-

1. The domain of the function $f(x) = \frac{\sqrt{\log_5(x-1)}}{\sqrt{4x+5-x^2}}$ is
(A) $x \in [2, 5)$ (B) $x \in (1, 5)$ (C) $x \in (-1, 5)$ (D) $x \in [4, 5)$
 2. If $f : [0, 3] \rightarrow [0, 5]$ is a bijective function defined by $f(x) = ax^2 + bx + 5$ (where $a, b \in \mathbb{R}$), then $f(3)$ equals
(A) 1 (B) 2 (C) 3 (D) 0
 3. If orthocentre and centroid of triangle are (2, 1) and $\left(\frac{8}{3}, \frac{11}{3}\right)$, then the co-ordinates of circumcentre are
(A) (2, 6) (B) (3, 5) (C) (2, 5) (D) (1, 4)
-

Space for Rough Work

4. If the domain of $g(x)$ is $[0, \infty)$ then the domain of $g(6\{x\} - 8\{x\}^2 - 1)$ is
 ($\{.\}$ denotes fractional part function)

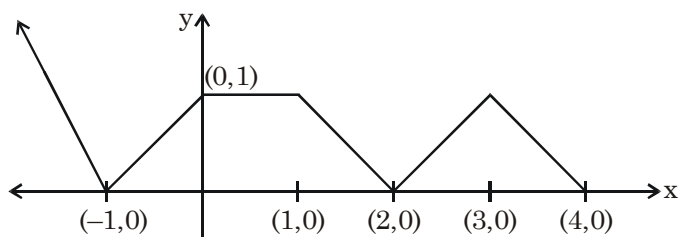
(A) $\bigcup_{n \in \mathbb{I}} \left(n + \frac{1}{4}, n + \frac{1}{2} \right)$

(B) $\bigcup_{n \in \mathbb{I}} \left[n + \frac{1}{6}, n + \frac{1}{2} \right]$

(C) $\bigcup_{n \in \mathbb{I}} \left[n + \frac{1}{4}, n + \frac{1}{2} \right]$

(D) $\bigcup_{n \in \mathbb{I}} \left(n + \frac{1}{6}, n + \frac{1}{2} \right)$

5. Given the graph of $y = f(x)$



Number of solutions of the equation $f(-|x|) = \frac{1}{2}$

- (A) 2 (B) 0 (C) 4 (D) 8
6. The range of the function $f(x) = 4|\cos x| - 3|\sin x|$ is
 (A) $[-3, 4]$ (B) $[-4, 3]$ (C) $(-5, 5)$ (D) $[-5, 5]$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. If $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x) = \begin{cases} x^2 - 9 & ; x \in \mathbb{Q} \\ x^3 - \sqrt{5} & ; x \notin \mathbb{Q} \end{cases}$, then $f(x)$ is

(where $\mathbb{Q}$ is the set of rational numbers)

- (A) one-one function (B) many-one function
 (C) into function (D) onto function

Space for Rough Work

8. Which of the following functions is/are an odd function ?

(A) $f(x) = \log(x + \sqrt{1+x^2})$ (B) $f(x) = x \left(\frac{e^x - 1}{e^x + 1} \right)$

(C) $f(x) = (x^3 + x) |\sin x|$ (D) $f(x) = 2$

9. The function $f : X \rightarrow Y$, defined by $f(x) = x^2 - 4x + 5$ is bijective function if

(A) $X = [2, \infty)$ and $Y = [1, \infty)$ (B) $X = (-\infty, 1)$ and $Y = (1, \infty)$

(C) $X = (-\infty, 1)$ and $Y = (2, \infty)$ (D) $X = [2, \infty)$ and $Y = [0, \infty)$

10. Period of the function $f(x) = [\sin 6x] + |\cos 12x|$ equals

(Where $[.]$ denotes greatest integer function)

(A) $\frac{2\pi}{3}$ (B) $\frac{\pi}{12}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{3}$

11. Which of the following pair of functions are **NOT** identical ?

(A) $f(x) = \sqrt{x^2}$ and $g(x) = (\sqrt{x})^2$ (B) $f(x) = \sqrt{\frac{x-2}{x-3}}$ and $g(x) = \frac{\sqrt{x-2}}{\sqrt{x-3}}$

(C) $f(x) = \tan x$ and $g(x) = \frac{1}{\cot x}$ (D) $f(x) = \frac{|x|}{x}$ and $g(x) = \operatorname{sgn}(x)$

Space for Rough Work

12. Consider the lines $L_1 : 3x + 4y = 10$ and $L_2 : 5x - 12y = 7$ then
 (A) the equation of bisector between L_1 and L_2 containing origin is $64x - 8y - 165 = 0$
 (B) the equation of bisector between L_1 and L_2 containing $(1, 2)$ is $64x - 8y - 165 = 0$
 (C) the equation of acute angle bisector between L_1 and L_2 is $14x + 112y - 95 = 0$
 (D) the equation of obtuse angle bisector between L_1 and L_2 is $64x - 8y - 165 = 0$
13. The equation of the line passing through the intersection of $7x + y = 1$ and $3x - 4y + 5 = 0$ and
 (A) which is parallel to the bisector of 2nd and 4th quadrant is $x + y = \frac{37}{31}$
 (B) which is parallel to the x-axis is $y = \frac{35}{31}$
 (C) which passes through $(1, 1)$ is $7x + 32y = 39$
 (D) which is parallel to the bisector of 2nd and 4th quadrant is $x + y = \frac{43}{31}$
14. The ΔABC is formed by the lines $2x - 3y = 6$, $3x - y + 6 = 0$ and $3x + 4y = 24$. If the point $P(\alpha, 0)$ and $Q(0, \beta)$ always lie on or inside the ΔABC then (Where $\alpha, \beta \in \mathbb{R}$)
 (A) $\alpha \in [-2, 3]$ (B) $\beta \in [-1, 6]$ (C) $\alpha \in [-2, 2]$ (D) $\beta \in [-2, 6]$

Space for Rough Work

SECTION-I(iii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

Paragraph for Questions 15 and 16

Consider the function $f(x) = 7x - x^2 - 10$.

15. The domain of the function $g(x) = \log_e f(x)$
(A) $x \in (2, 5)$ (B) $x \in [2, 5]$ (C) $x \in (1, 6)$ (D) $x \in (3, 4)$
16. The range of the function $g(x) = \log_e f(x)$
(A) $\left(-\infty, \log_e \frac{13}{4}\right]$ (B) $\left(-\infty, \log_e \frac{9}{4}\right]$ (C) $(-\infty, \log_e 2]$ (D) $\left[\log_e \frac{13}{4}, \infty\right]$

Space for Rough Work

Paragraph for Questions 17 and 18

Consider the function $f(x) = -1 + |x - 2|$, $0 \leq x \leq 4$

$$g(x) = 2 - |x|, -1 \leq x \leq 3$$

17. Domain of $f(g(x))$ is

(A) $x \in [-1, 3]$

(B) $x \in [0, 4]$

(C) $x \in [-1, 2]$

(D) $x \in [1, 4]$

18. Range of $f(g(x))$ is

(A) $[0, 2]$

(B) $[1, 2]$

(C) $[0, 3]$

(D) $[-1, 1]$

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has one sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	Single correct option	6	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	–1 In all other cases	18
I(ii)	One or more correct option(s)	8	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	–2 In all other cases	32
I(iii)	Paragraph Based (Single correct option)	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 In all other cases	—	12

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator